

EXPERIMENT 13

IMPLEMENTATION OF DIFFIE-HELLMAN

AIM

To implement the Diffie-Hellman algorithm for secure key exchange between two users.

ALGORITHM

1. Choose a prime number (P) and generator (G).
2. Enter the private keys of User A and User B.
3. Generate the public keys.
4. Exchange the public keys.
5. Compute the shared secret key.
6. Display the shared keys.

CODE

```
#include <stdio.h>

long long power(long long a,long long b,long long m)
{
    long long r=1;
    while(b>0)
    {
        r=(r*a)%m;
        b--;
    }
    return r;
}

int main()
{
    int P=23,G=5,a,b;
    long long A,B,key1,key2;

    printf("Enter Private Key of User A: ");
    scanf("%d",&a);

    printf("Enter Private Key of User B: ");
    scanf("%d",&b);

    A=power(G,a,P);
```

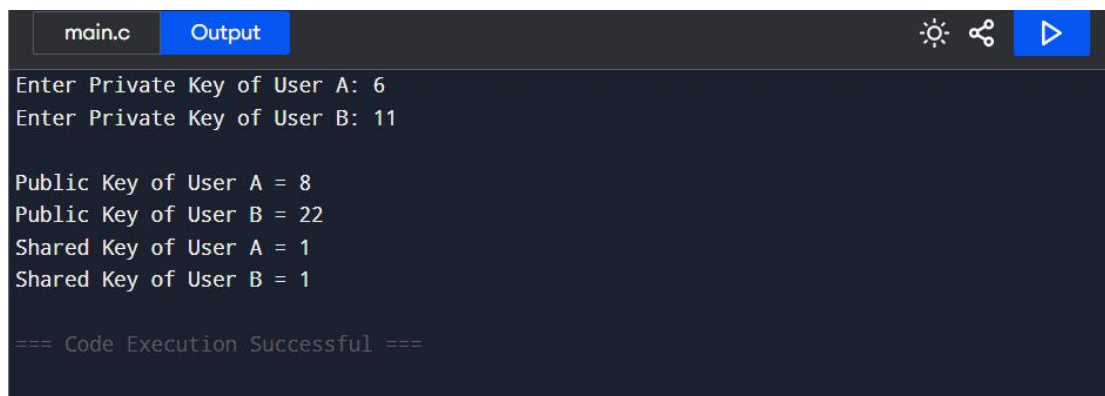
```
B=power(G,b,P);

key1=power(B,a,P);
key2=power(A,b,P);

printf("\nPublic Key of User A = %lld",A);
printf("\nPublic Key of User B = %lld",B);
printf("\nShared Key of User A = %lld",key1);
printf("\nShared Key of User B = %lld",key2);

return 0;
}
```

OUTPUT



The screenshot shows a code editor with a dark theme. The top bar has tabs for 'main.c' and 'Output', with 'Output' selected. On the right of the top bar are icons for settings, a share icon, and a play button. The main area displays the program's output in a monospaced font. It shows the user entering private keys for User A (6) and User B (11). The program then outputs the calculated public keys (8 for A, 22 for B) and the shared secret key (1 for both). At the bottom, it states '=== Code Execution Successful ==='.

```
main.c  Output  [Settings] [Share] [Run]

Enter Private Key of User A: 6
Enter Private Key of User B: 11

Public Key of User A = 8
Public Key of User B = 22
Shared Key of User A = 1
Shared Key of User B = 1

=== Code Execution Successful ===
```

RESULT

The Diffie-Hellman algorithm successfully generated the same shared secret key for both users.